

An extended generalized Pareto regression model for count data

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Collaboration

- This work is done in collaboration with:



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- ① Motivation
- ② Introduction & background
- ③ Flexible model construction
- ④ Applications

1 Motivation

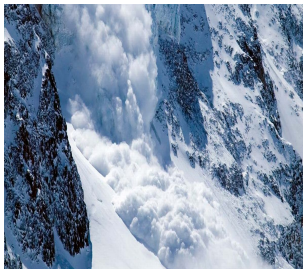
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Motivation: extremes of avalanches

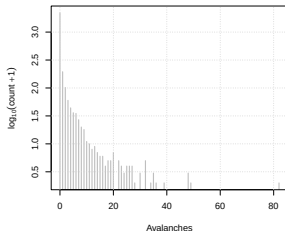
- **Snow avalanches** are among the most destructive natural hazards in mountainous regions and threatening to:
 - human lives
 - ecosystems
 - built structures and
 - landscapes



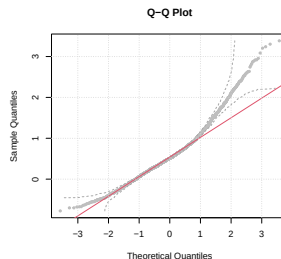
<https://www.sudouest.fr/tourisme/pyrenees/meteo-france-alerte-sur-les-avalanches-3531479.php>.

Motivation: fit of existing model

- Zero-inflated negative binomial regression model fitted to avalanche counts of Haute-Maurienne massif.



(a)



(b)

Motivation: encounter the problem

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- Predicting the occurrence and measuring the risk of extreme avalanches is a longstanding issue.
- An efficient modeling framework is required when we have **count** data of avalanches with **extreme observations**.
- An extreme value model, or a heavy-tailed distribution, may be helpful for such extreme events modeling.

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Role of Extreme Value Theory

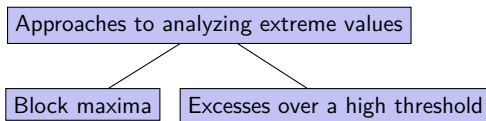
⇒ What is extreme event?

An **extreme event** is a rare, unusual, and severe occurrence that significantly deviates from normal conditions and substantially impacts the environment, society, or economy. (**Hurricanes, heatwaves, earthquakes, and extreme precipitations, droughts.**)

⇒ What is extreme value theory?

Extreme Value Theory (EVT) is a statistical approach that focuses on **analyzing extreme events** or values in data rather than assuming a normal or symmetric distribution.

Role of Extreme Value Theory: Main approaches



- The **block maxima** approach reduces the data considerably by taking **maxima of long blocks of data**, e.g., annual maxima. The **Generalized Extreme Value** distribution has theoretical justification for fitting to block maxima of data (Coles, 2001).
- The **Generalized Pareto distribution (GPD)** function has a similar justification for fitting to excesses over a high threshold.

Role of Extreme Value Theory: Discrete extremes

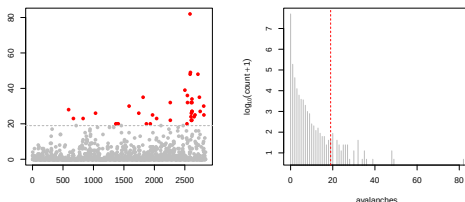
⇒ Discrete extremes

- On possibility is to model threshold exceedances using **discrete generalized Pareto distribution** (DGPD) Hitz et al. (2017).

The DGPD is defined by discretizing

$$\mathcal{F}(k) = 1 - (1 + k\xi/\sigma)^{-1/\xi}$$

$$\Pr(Y-u = k | Y \geq u) = \mathcal{F}(k+1; \sigma, \xi) - \mathcal{F}(k; \sigma, \xi), \quad k \in \mathbb{N}_0 \quad (1)$$



Discrete GPD

⇒ **Why not using continuous GPD...?**

- The continuous maximum domain of attraction is preserved under discretization for $\xi > 0$ (Shimura, 2012; Hitz et al., 2017)

Discrete GPD

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- The continuous maximum domain of attraction is preserved under discretization for $\xi > 0$ (Shimura, 2012; Hitz et al., 2017)
- Considering discrete exceedances as continuous in the presence of ties leads to bias in estimation (Hitz et al., 2017).

Problems with DGPD fitting

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- Optimal threshold selection is problematic.
- Losing a large amount of data when using high quantile for threshold selection.
- We required a **flexible model** that accurately models count data across different scenarios:
 - ① the entire data with a heavy tail
 - ② the data with excessive zeros and heavier tail together
 - ③ the data above a very low threshold (Ahmad and Arshad, 2024)

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Discrete extended generalized Pareto distribution (DEGDP)

- Construction is based on the integral transform idea to simulate GPD random draws Naveau et al. (2016), that is

$$\mathcal{F}_{\sigma,\xi}^{-1}(U), \quad U \sim \mathcal{U}(0, 1)$$

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- This leads to the class of random variables stochastically defined as,

$$\mathcal{F}_{\sigma,\xi}^{-1}\{G^{-1}(U)\}, \quad (2)$$

where G is a CDF on $[0, 1]$ and $U \sim \mathcal{U}(0, 1)$.

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- The **key problem** is to find a class for G which preserves the upper tail behavior with shape parameter ξ and also controls the lower tail behavior

DEGPD

- Naveau et al. (2016) defined restrictions for validity of G families. For instance, the tail of G denoted by $\bar{G} = 1 - G$ has to satisfy

$$\lim_{u \rightarrow 0} \frac{\bar{G}(1-u)}{u} = a, \text{ for some finite } a > 0 \text{ (upper tail behavior),}$$
$$\lim_{u \rightarrow 0} \frac{G(u)}{u^\kappa} = c, \text{ for some finite } c > 0 \text{ (lower tail behavior).} \quad (3)$$

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- We start by considering the CDF $G\{\mathcal{F}(\cdot; \sigma, \xi)\}$ of EGPD where G meets the conditions (3). To model non-negative integer data, we discretize the CDF by

$$\Pr(Y = k) = G\{\mathcal{F}(k+1; \sigma, \xi)\} - G\{\mathcal{F}(k; \sigma, \xi)\}, \quad k \in \mathbb{N}_0 \quad (4)$$

The distribution will be called **DEGPD**.

Zero-inflated DEGPD

⇒ **Zero inflation problem along with heavier tail...**

Let Z be observed with an **excessive number of zeros** relative to those observed under the DEGPD. The zero-inflated distribution (**ZIDEGPD**) is defined in a straightforward way as:

$$\Pr(Z = m) = \begin{cases} \pi + (1 - \pi)G\{\mathcal{F}(1, \sigma, \xi)\} & m = 0 \\ (1 - \pi)[G\{\mathcal{F}(m + 1, \sigma, \xi)\} - G\{\mathcal{F}(m, \sigma, \xi)\}] & m \in \mathbb{N} \end{cases} \quad (5)$$

where $0 \leq \pi \leq 1$ and the remaining parameters would be the same as DEGPD.

It turns out that the CDF of ZIDEGPD is

$$\Pr(Z \leq m) = \pi + (1 - \pi)G\{\mathcal{F}(m + 1, \sigma, \xi)\}, \quad m \in \mathbb{N}_0 \quad (6)$$

G families

- For G , we use four parametric expressions $G(\cdot, \psi)$, already proposed in Naveau et al. (2016), which preserve the upper tail behavior with shape parameter ξ .

- $G(u; \psi) = u^\kappa, \quad \psi = \kappa > 0$

- $G(u; \psi) = 1 - D_\delta \left\{ (1 - u)^\delta \right\} \quad \psi = \delta > 0$

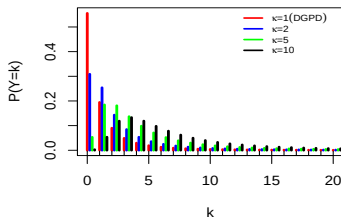
where D_δ is the CDF of beta distribution with parameters $1/\delta$ and 2.

- $G(u; \psi) = \left[1 - D_\delta \left\{ (1 - u)^\delta \right\} \right]^{\kappa/2} \quad \psi = (\delta, \kappa) > 0$

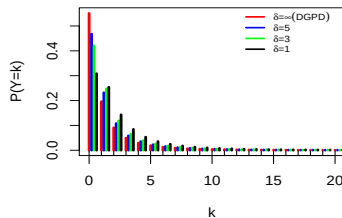
- $G(u; \psi) = pv^{\kappa_1} + (1 - p)v^{\kappa_2} \quad \psi = (p, \kappa_1, \kappa_2), \kappa_1, \kappa_2 > 0 \text{ and } p \in [0, 1], \kappa_1 \leq \kappa_2$

PMF behaviors of DEGPD

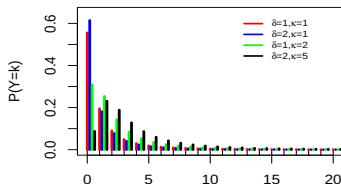
Model type (i) $G(u;\psi)=u^{\kappa}$



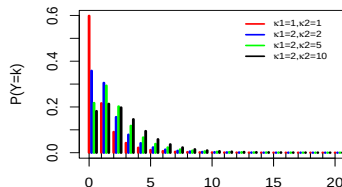
Model type (ii) $G(u;\psi)=1-D_{\delta}((1-u)^{\delta})$



Model type (iii) $G(u;\psi)=(1-D_{\delta}((1-u)^{\delta}))^{\kappa/2}$

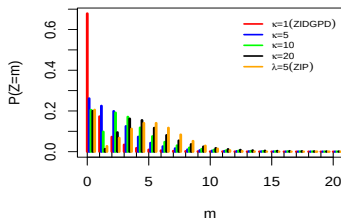


Model type (vi) $G(u;\psi)=pu^{\kappa_1}+(1-p)u^{\kappa_2}$

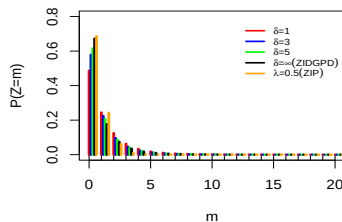


PMF behaviour of ZIDEGPD

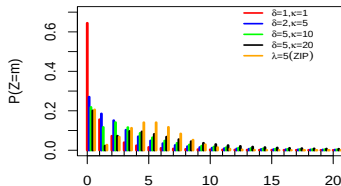
Model type (i) $G(v)=v^{\kappa}$



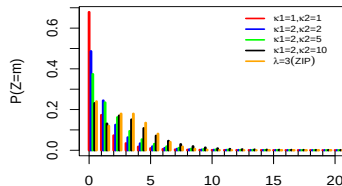
Model type (ii) $G(v)=1-F_{\delta}((1-v)^{\delta})$



Model type (iii) $G(v)=(1-F_{\delta}((1-v)^{\delta}))^{\kappa/2}$



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Generalized additive modeling (GAM)

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- In the sequel, we denote the vector of parameters $(\pi, \xi, \sigma, \psi^T)^T$ with $\theta = (\theta_1, \dots, \theta_d)^T$
- The parameters of the distribution of Y may depend on some covariates $\mathbf{x}$, i.e. $\theta(\mathbf{x}) = (\theta_1(\mathbf{x}), \dots, \theta_d(\mathbf{x}))^T$.

Generalized additive modeling

- For relating the distributional parameters $(\theta_1(\mathbf{x}), \dots, \theta_d(\mathbf{x}))$ to the covariates, we consider additive predictors of the form

$$\eta_i(\mathbf{x}) = s_{i1}(\mathbf{x}) + \dots + s_{iJ_i}(\mathbf{x}) \quad (7)$$

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- The predictors are linked to the distributional parameters via known monotonic and twice differentiable link functions $h_i(\cdot)$.

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- In the case of model with $G(u; \psi) = u^\kappa$, common link functions are

$$\begin{aligned} \xi(\mathbf{x}) &= \exp(\eta_\xi(\mathbf{x})), & \sigma(\mathbf{x}) &= \exp(\eta_\sigma(\mathbf{x})), & \kappa(\mathbf{x}) &= \exp(\eta_\kappa(\mathbf{x})), \\ \pi(\mathbf{x}) &= \exp\left(\frac{\eta_\pi(\mathbf{x})}{1 + \eta_\pi(\mathbf{x})}\right) \end{aligned}$$

Generalized additive modeling

- The functions s_{ij} in (7) are approximated in terms of basis function expansions,

$$s_{ij}(\mathbf{x}) = \sum_{k=1}^{K_{ij}} \beta_{ij,k} B_k(\mathbf{x}), \quad (9)$$

where $B_k(\mathbf{x})$ are the basis functions and $\beta_{ij,k}$ denote the corresponding basis coefficients.

- The basis function expansions can be written as $s_{ij}(\mathbf{x}) = \mathbf{t}_{ij}(\mathbf{x})^T \boldsymbol{\beta}_{ij}$ where $\mathbf{t}_{ij}(\mathbf{x})$ is still a vector of transformed covariates that depends on the basis functions and $\boldsymbol{\beta}_{ij} = (\beta_{ij,1}, \dots, \beta_{ij,K_{ij}})^T$ is a parameter vector to be estimated.

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Estimation

- The likelihood function for ZIDEGPD (5) is written as.

$$\begin{aligned} l(\beta) = & \sum_{i=1}^n l_0(y_i) \log [\pi + (1 - \pi)G(\mathcal{F}(1; \sigma, \xi); \kappa)] \\ & + \sum_{i=1}^n (1 - l_0(y_i)) \log(1 - \pi) \times \left[G(\mathcal{F}(y_i + 1; \sigma, \xi); \kappa) \right. \\ & \left. - G(\mathcal{F}(y_i; \sigma, \xi); \kappa) \right], \end{aligned} \quad (10)$$

where β collects all unknown coefficient β_{ijk} of the basis expansions.

- The penalized log-likelihood function for the latter models reads:

$$l_p(\beta) = l(\beta) - \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^{J_i} \beta_{ij}^T \mathbf{G}_{ij}(\lambda_{ij}) \beta_{ij} \quad (11)$$

where $l(\beta)$ is the log-likelihood function (10).

- To fit ZIDEGPD with GAM forms, we have written an R code that implements the distributions as “new families” for `evgam` R package (Youngman, 2022).¹

¹An example of R code with the name “Fit_degpd_zidegpd.R” and the complete source code of the function is provided on the GitHub <https://github.com/touqeerahmadunipd/degpd-and-zidegpd>.

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Automobile insurance complaints data

Application to avalanches data

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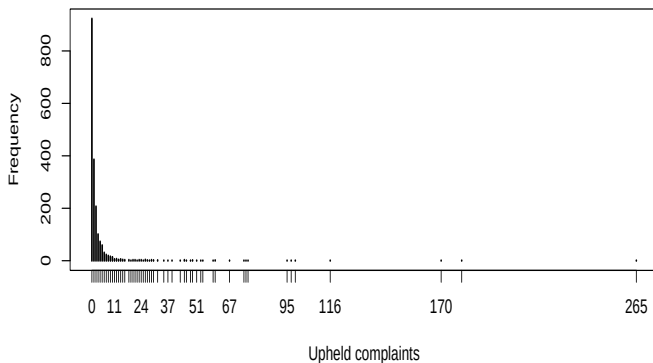
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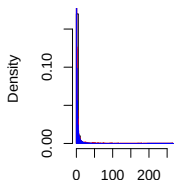
- Upheld complaints of auto insurance companies of New York city recorded from 2009 to 2020 ($n=1942$).



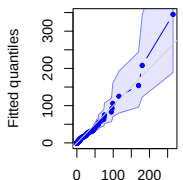
Application to auto insurance complaints data

Table 1: AIC and BIC associated with the fitted DEGPD and Chi-square goodness-of-fit test.

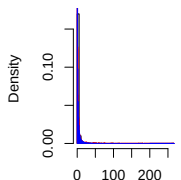
Model	DEGPD		
	AIC	BIC	Chi-square
(1)	7290.93	7307.65	0.20 (0.99)
(2)	7291.88	7308.60	0.19 (0.99)
(3)	7293.36	7315.64	0.20 (0.99)
(4)	7294.45	7322.30	0.20 (0.99)



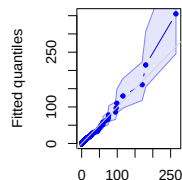
Upheld complaints



Empirical quantiles



Upheld complaints



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Application to avalanches data with environmental covariates

- We consider in particular **three-day observation window** of the daily number of avalanche events recorded from February 1982 to April 2021 (October 15th and May 15th). Further, we consider the following covariates

Name	Definition
WS	maximum wind speed at 10 meters (m/s)
PREC	precipitation (mm/day)
MxT	maximum temperature at 2 Meters (C°)
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- Three-day summaries are calculated by averaging RH and PREC and taking the maximum for MxT and WS and the minimum for MnT.

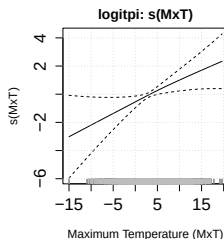
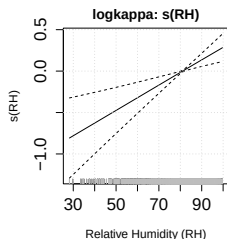
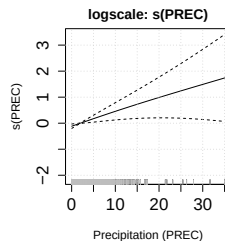
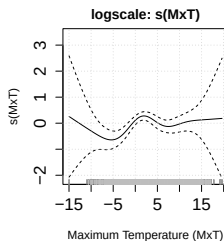
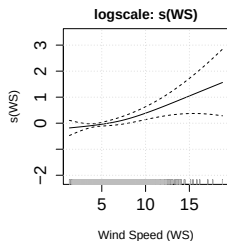
Selection of covariates

- After comparing different combinations of the covariates, we found WS, MxT, PREC, and RH are more appropriate to use as covariates.
- For instance, it turned out for the model $G(u, \Psi) = u^\kappa$ with the lowest AIC are

$$\begin{aligned} \text{Model type (1)} : \quad \kappa &= s(RH); \quad \sigma = s(WS) + s(MxT) \\ &+ s(PREC); \quad \xi = cst \\ \text{logit}(\pi) &= s(MxT) \end{aligned}$$

where $s(\cdot)$ indicates the smoothed predictor.

Non-parametric effects of covariates



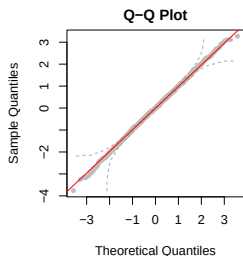
Model adequacy

- To assess the proposed model and compare it with the existing models, we use randomized residuals analysis (Dunn and Smyth, 1996).

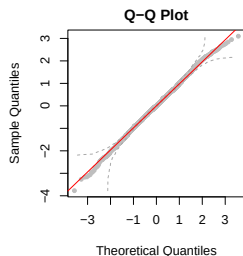
$$r_i = \Phi^{-1}(\nu_i)$$

where Φ is a standard normal distribution function,
 $\nu_i = (1 - u_i)F(m_i - 1; \hat{\theta}_i) + u_i F(m_i; \hat{\theta}_i)$, u_i is drawn from a uniform distribution and $F(., \hat{\theta})$ is the parametric estimate of the CDF of the current models defined in (6).

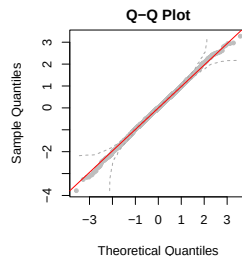
Model adequacy



Model (i)



Model (ii)



Model (iii)

Goodness-of-fit

Table 2: AIC values and Anderson-Darling normality test statistics of the different ZIDEGPD models.

Model	AIC	Statistics	P-value
(i)	6121	0.287	0.623
(ii)	6151	0.367	0.432
(iii)	6152	0.315	0.543

Importance and further applications

- Motivated by the avalanche real data example, we attempt to properly model the lower limit of the distribution and the upper tail without neglecting the bulk part of the data.

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- Non-additive models can be fitted using thin-plate splines for isotropic spatial dependence or tensor product splines for covariate interactions.
- One can propose a new model if the new G-family fulfills the condition (3).

Ahmad, T. and Arshad, I. A. (2024). New flexible versions of extended generalized pareto model for count data. *arXiv preprint arXiv:2409.18719*.

Coles (2001). *An introduction to statistical modeling of extreme values*, volume 208. Springer.

Dunn, P. K. and Smyth, G. K. (1996). Randomized quantile residuals. *Journal of Computational and graphical statistics*, 5(3):236–244.

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Thank you for your attention!

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